Phân tích về Genome Island, Pathogenicity Island và HGT của mẫu B13

Genome Island

Giới thiệu về GI

- Genome Island là các vùng DNA có kích thước lớn (10-200 kB)
- Được thu nhận từ bên ngoài vào tế bào vi khuẩn bằng HGT
- Có sự khác biệt rõ rệt so với genome của chính vi khuẩn gốc

Đặc điểm của GI

- Có **GC content** và **Codon usage** khác biệt so với hệ gene gốc
- Mang gene của **mobile genetic elements** (integrase, transpose, recombinase)
- Thường chèn gần tRNA
- Mang các gene có chức năng về độc lực, kháng kháng sinh, ...

Phân tích GI

IslandViewer 4

- Tích hợp 3 thuật toán: IslandPick (comparative genomics), SIGI-HMM (codon usage bias), IslandPath-DIMOB (GC + mobility genes)
- GI được dự đoán khi ≥1 phương pháp đồng thuận
- Kết quả bao gồm: Danh sách GI (Vị trí, kích thước, các gene trong GI) + Visualization
- Phụ thuộc dữ liệu tham chiếu (Bacillus subtilis str. 168)

Island start	Island end	Length	Gene name	Gene ID	Locus	Gene start	Gene end	Strand	Product
489050	493112	4062		yjeH_1	OGCDPPDJ_03694	489050	490309	-1	L-methionine/branched-chain amino acid exporter YjeH
489050	493112	4062		gltR_5	OGCDPPDJ_03693	490440	491339	1	HTH-type transcriptional regulator GltR
489050	493112	4062			OGCDPPDJ_03692	491832	492395	-1	hypothetical protein
489050	493112	4062			OGCDPPDJ_03691	492594	493112	-1	hypothetical protein
670760	676918	6158		rlmCD_1	OGCDPPDJ_04116	670760	672136	1	23S rRNA (uracil-C(5))-methyltransferase RlmCD
670760	676918	6158			OGCDPPDJ_04117	673459	673668	-1	hypothetical protein
670760	676918	6158			OGCDPPDJ_04118	673901	676918	1	hypothetical protein
726747	730932	4185			OGCDPPDJ_04173	726747	727832	1	hypothetical protein
726747	730932	4185			OGCDPPDJ_04174	727837	729747	1	hypothetical protein
726747	730932	4185			OGCDPPDJ_04175	729871	730932	-1	hypothetical protein
1055380	1059868	4488		glcP	OGCDPPDJ_03418	1055380	1056585	-1	Glucose/mannose transporter GlcP
1055380	1059868	4488		ntdC	OGCDPPDJ_03417	1056657	1057709	-1	Glucose-6-phosphate 3-dehydrogenase
1055380	1059868	4488		ntdB	OGCDPPDJ_03416	1057723	1058571	-1	Kanosamine-6-phosphate phosphatase
1055380	1059868	4488		ntdA	OGCDPPDJ_03415	1058543	1059868	-1	3-oxo-glucose-6-phosphate:glutamate aminotransferase
1191680	1270652	78972			OGCDPPDJ_03276	1191680	1192870	-1	hypothetical protein
1191680	1270652	78972		xerC_6	OGCDPPDJ_03274	1193272	1194315	1	Tyrosine recombinase XerC
1191680	1270652	78972			OGCDPPDJ_03273	1194494	1195456	1	hypothetical protein

Island start	Island end	Length	Gene name	Gene ID	Locus	Gene start	Gene end	Strand	Product
1191680	1270652	78972			OGCDPPDJ_03272	1195453	1195872	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03271	1196321	1196518	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03270	1196515	1196877	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03269	1196877	1197011	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03268	1197008	1197385	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03267	1197414	1198166	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03266	1198163	1198315	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03265	1198335	1199117	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03264	1199114	1199632	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03263	1199632	1200012	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03262	1200009	1201397	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03261	1201507	1202520	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03260	1202810	1203019	-1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03259	1203122	1204291	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03258	1204233	1204355	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03257	1204585	1205112	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03256	1205253	1205408	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03255	1205405	1205971	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03254	1205964	1206071	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03253	1206219	1207034	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03252	1207077	1207499	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03251	1207499	1207744	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03250	1207756	1209021	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03249	1209187	1209339	1	hypothetical protein
1191680	1270652	78972		polA_2	OGCDPPDJ_03248	1209374	1211659	1	DNA polymerase I
1191680	1270652	78972		dinG_3	OGCDPPDJ_03247	1211661	1212365	1	3'-5' exonuclease DinG
1191680	1270652	78972			OGCDPPDJ_03246	1212366	1213424	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03245	1213424	1213621	1	hypothetical protein
1191680	1270652	78972		ruvC	OGCDPPDJ_03244	1213618	1214178	1	Crossover junction endodeoxyribonuclease RuvC
1191680	1270652	78972			OGCDPPDJ_03243	1214179	1214541	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03242	1214547	1214741	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03241	1214738	1214896	1	hypothetical protein
1191680	1270652	78972		nrdI_3	OGCDPPDJ_03240	1214890	1215252	1	Protein NrdI
1191680	1270652	78972		nrdE_5	OGCDPPDJ_03239	1215259	1217346	1	Ribonucleoside-diphosphate reductase subunit alpha
1191680	1270652	78972			OGCDPPDJ_03238	1217366	1217542	1	hypothetical protein
1191680	1270652	78972		nrdF_3	OGCDPPDJ_03237	1217556	1218539	1	Ribonucleoside-diphosphate reductase subunit beta
1191680	1270652	78972			OGCDPPDJ_03236	1218636	1218794	1	hypothetical protein

Island start	Island end	Length	Gene name	Gene ID	Locus	Gene start	Gene end	Strand	Product
1191680	1270652	78972			OGCDPPDJ_03235	1218791	1219183	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03234	1219200	1219691	1	hypothetical protein
1191680	1270652	78972		thyX	OGCDPPDJ_03233	1219692	1220504	1	Flavin-dependent thymidylate synthase
1191680	1270652	78972			OGCDPPDJ_03232	1220531	1221097	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03231	1221115	1221615	1	Protein SprT-like protein
1191680	1270652	78972			OGCDPPDJ_03230	1221616	1222353	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03229	1222350	1222592	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03228	1222589	1223170	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03227	1223143	1223523	-1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03226	1223619	1223987	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03225	1223987	1224484	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03224	1224523	1224702	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03223	1224734	1225162	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03222	1225215	1225400	1	hypothetical protein
1191680	1270652	78972		lig_2	OGCDPPDJ_03221	1225437	1226246	-1	DNA ligase
1191680	1270652	78972			OGCDPPDJ_03220	1226340	1226759	-1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03219	1226777	1226959	-1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03218	1227137	1227574	-1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03217	1227571	1227687	-1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03216	1227680	1227883	-1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03215	1227888	1227977	-1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03214	1228245	1230011	-1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03213	1230011	1230808	-1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03212	1230940	1231161	1	hypothetical protein
1191680	1270652	78972		epsH_2	OGCDPPDJ_03211	1231260	1232282	1	Putative glycosyltransferase EpsH
1191680	1270652	78972			OGCDPPDJ_03210	1232641	1233219	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03209	1233343	1233519	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03208	1233516	1233686	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03207	1233833	1234033	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03206	1234053	1234352	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03205	1234349	1234570	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03204	1234567	1234752	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03203	1234749	1235045	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03202	1235066	1235890	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03201	1236146	1236514	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03200	1236471	1236845	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03199	1236967	1237290	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03198	1237265	1239046	1	hypothetical protein

Island start	Island end	Length	Gene name	Gene ID	Locus	Gene start	Gene end	Strand	Product
1191680	1270652	78972			OGCDPPDJ_03197	1239058	1240356	1	hypothetical protein
1191680	1270652	78972		clpP_2	OGCDPPDJ_03196	1240310	1241056	1	ATP-dependent Clp protease proteolytic subunit
1191680	1270652	78972			OGCDPPDJ_03195	1241056	1242210	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03194	1242251	1242739	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03193	1242742	1242981	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03192	1242986	1243327	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03191	1243324	1243752	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03190	1243749	1244093	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03189	1244096	1244698	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03188	1244781	1245104	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03187	1245332	1250803	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03186	1250803	1251621	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03185	1251631	1253157	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03184	1253175	1255223	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03183	1255242	1256465	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03182	1256480	1256773	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03181	1256774	1256944	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03180	1256967	1257233	1	hypothetical protein
1191680	1270652	78972		cwlA_4	OGCDPPDJ_03179	1257302	1258243	1	N-acetylmuramoyl-L-alanine amidase CwlA
1191680	1270652	78972			OGCDPPDJ_03178	1258262	1258519	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03177	1258641	1258961	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03176	1258958	1259293	-1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03175	1259339	1259563	-1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03174	1259833	1260120	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03173	1260117	1261271	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03172	1261258	1261887	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03171	1261884	1262078	-1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03170	1262205	1263032	1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03169	1263476	1263676	-1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03168	1263730	1264038	-1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03167	1264059	1265942	-1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03166	1266217	1266453	-1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03165	1266507	1267235	-1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03164	1267452	1267664	-1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03163	1268363	1268878	-1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03162	1269007	1269207	-1	hypothetical protein
1191680	1270652	78972			OGCDPPDJ_03161	1269711	1269902	-1	hypothetical protein
1191680	1270652	78972		yjcS	OGCDPPDJ_03160	1270371	1270652	1	putative protein YjcS

Island start	Island end	Length	Gene name	Gene ID	Locus	Gene start	Gene end	Strand	Product
1217556	1224484	6928		nrdF_3	OGCDPPDJ_03237	1217556	1218539	1	Ribonucleoside-diphosphate reductase subunit beta
1217556	1224484	6928			OGCDPPDJ_03236	1218636	1218794	1	hypothetical protein
1217556	1224484	6928			OGCDPPDJ_03235	1218791	1219183	1	hypothetical protein
1217556	1224484	6928			OGCDPPDJ_03234	1219200	1219691	1	hypothetical protein
1217556	1224484	6928		thyX	OGCDPPDJ_03233	1219692	1220504	1	Flavin-dependent thymidylate synthase
1217556	1224484	6928			OGCDPPDJ_03232	1220531	1221097	1	hypothetical protein
1217556	1224484	6928			OGCDPPDJ_03231	1221115	1221615	1	Protein SprT-like protein
1217556	1224484	6928			OGCDPPDJ_03230	1221616	1222353	1	hypothetical protein
1217556	1224484	6928			OGCDPPDJ_03229	1222350	1222592	1	hypothetical protein
1217556	1224484	6928			OGCDPPDJ_03228	1222589	1223170	1	hypothetical protein
1217556	1224484	6928			OGCDPPDJ_03227	1223143	1223523	-1	hypothetical protein
1217556	1224484	6928			OGCDPPDJ_03226	1223619	1223987	1	hypothetical protein
1217556	1224484	6928			OGCDPPDJ_03225	1223987	1224484	1	hypothetical protein
1227888	1232282	4394			OGCDPPDJ_03215	1227888	1227977	-1	hypothetical protein
1227888	1232282	4394			OGCDPPDJ_03214	1228245	1230011	-1	hypothetical protein
1227888	1232282	4394			OGCDPPDJ_03213	1230011	1230808	-1	hypothetical protein
1227888	1232282	4394			OGCDPPDJ_03212	1230940	1231161	1	hypothetical protein
1227888	1232282	4394		epsH_2	OGCDPPDJ_03211	1231260	1232282	1	Putative glycosyltransferase EpsH
1233833	1239046	5213			OGCDPPDJ_03207	1233833	1234033	1	hypothetical protein
1233833	1239046	5213			OGCDPPDJ_03206	1234053	1234352	1	hypothetical protein
1233833	1239046	5213			OGCDPPDJ_03205	1234349	1234570	1	hypothetical protein
1233833	1239046	5213			OGCDPPDJ_03204	1234567	1234752	1	hypothetical protein
1233833	1239046	5213			OGCDPPDJ_03203	1234749	1235045	1	hypothetical protein
1233833	1239046	5213			OGCDPPDJ_03202	1235066	1235890	1	hypothetical protein
1233833	1239046	5213			OGCDPPDJ_03201	1236146	1236514	1	hypothetical protein
1233833	1239046	5213			OGCDPPDJ_03200	1236471	1236845	1	hypothetical protein
1233833	1239046	5213			OGCDPPDJ_03199	1236967	1237290	1	hypothetical protein
1233833	1239046	5213			OGCDPPDJ_03198	1237265	1239046	1	hypothetical protein
2072966	2077007	4041			OGCDPPDJ_01271	2072966	2073304	1	hypothetical protein
2072966	2077007	4041			OGCDPPDJ_01270	2073764	2074048	-1	hypothetical protein
2072966	2077007	4041			OGCDPPDJ_01269	2074082	2074411	-1	hypothetical protein
2072966	2077007	4041			OGCDPPDJ_01268	2075357	2075563	-1	hypothetical protein
2072966	2077007	4041			OGCDPPDJ_01267	2075573	2075824	-1	hypothetical protein
2072966	2077007	4041			OGCDPPDJ_01266	2076090	2077007	-1	hypothetical protein
2077408	2085537	8129			OGCDPPDJ_01264	2077408	2077800	-1	hypothetical protein
2077408	2085537	8129			OGCDPPDJ_01263	2077845	2078066	-1	hypothetical protein
2077408	2085537	8129			OGCDPPDJ_01262	2078100	2078648	-1	hypothetical protein

Island start	Island end	Length	Gene name	Gene ID	Locus	Gene start	Gene end	Strand	Product
2077408	2085537	8129			OGCDPPDJ_01261	2078680	2079045	-1	hypothetical protein
2077408	2085537	8129			OGCDPPDJ_01260	2079276	2080079	-1	hypothetical protein
2077408	2085537	8129			OGCDPPDJ_01259	2080206	2081144	-1	hypothetical protein
2077408	2085537	8129			OGCDPPDJ_01258	2081575	2082084	-1	hypothetical protein
2077408	2085537	8129			OGCDPPDJ_01257	2082138	2082362	-1	hypothetical protein
2077408	2085537	8129			OGCDPPDJ_01256	2082440	2082586	-1	hypothetical protein
2077408	2085537	8129			OGCDPPDJ_01255	2082587	2082835	-1	hypothetical protein
2077408	2085537	8129			OGCDPPDJ_01254	2082828	2083001	-1	hypothetical protein
2077408	2085537	8129			OGCDPPDJ_01253	2083173	2083382	-1	hypothetical protein
2077408	2085537	8129			OGCDPPDJ_01252	2083862	2083987	-1	hypothetical protein
2077408	2085537	8129		rapG_1	OGCDPPDJ_01251	2084422	2085537	1	Response regulator aspartate phosphatase G
2077408	2085537	8129			OGCDPPDJ_01250	2085534	2085656	1	hypothetical protein
2148724	2163692	14968		pgpB	OGCDPPDJ_01176	2148724	2149335	-1	Phosphatidylglycerophosphatase B
2148724	2163692	14968			OGCDPPDJ_01175	2149412	2149588	-1	hypothetical protein
2148724	2163692	14968			OGCDPPDJ_01174	2149707	2149850	1	hypothetical protein
2148724	2163692	14968			OGCDPPDJ_01173	2149847	2150527	-1	hypothetical protein
2148724	2163692	14968			OGCDPPDJ_01172	2150675	2150899	-1	hypothetical protein
2148724	2163692	14968			OGCDPPDJ_01171	2150986	2151264	-1	hypothetical protein
2148724	2163692	14968		kamA_2	OGCDPPDJ_01170	2151261	2151710	-1	L-lysine 2,3-aminomutase
2148724	2163692	14968			OGCDPPDJ_01169	2151720	2151833	-1	hypothetical protein
2148724	2163692	14968			OGCDPPDJ_01168	2152122	2152403	1	hypothetical protein
2148724	2163692	14968			OGCDPPDJ_01167	2152403	2153026	1	hypothetical protein
2148724	2163692	14968			OGCDPPDJ_01166	2153058	2153294	1	hypothetical protein
2148724	2163692	14968			OGCDPPDJ_01165	2153337	2153555	1	hypothetical protein
2148724	2163692	14968			OGCDPPDJ_01164	2153574	2153954	1	hypothetical protein
2148724	2163692	14968			OGCDPPDJ_01163	2154257	2154427	-1	hypothetical protein
2148724	2163692	14968			OGCDPPDJ_01162	2155091	2155381	-1	hypothetical protein
2148724	2163692	14968		bamHIM	OGCDPPDJ_01161	2155804	2157120	-1	Modification methylase BamHI
2148724	2163692	14968			OGCDPPDJ_01160	2157192	2157464	1	hypothetical protein
2148724	2163692	14968		bamHIR	OGCDPPDJ_01159	2157454	2158095	1	Type-2 restriction enzyme BamHI
2148724	2163692	14968		xerD_2	OGCDPPDJ_01158	2158710	2159276	1	Tyrosine recombinase XerD
2148724	2163692	14968			OGCDPPDJ_01157	2159706	2160368	-1	hypothetical protein
2148724	2163692	14968			OGCDPPDJ_01156	2161058	2162620	-1	hypothetical protein
2148724	2163692	14968		kamA_1	OGCDPPDJ_01155	2162673	2163692	-1	L-lysine 2,3-aminomutase
2175080	2297014	121934			OGCDPPDJ_01140	2175080	2175670	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01139	2175706	2175987	1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01138	2175989	2176120	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01137	2176225	2176659	-1	hypothetical protein

Island start	Island end	Length	Gene name	Gene ID	Locus	Gene start	Gene end	Strand	Product
2175080	2297014	121934			OGCDPPDJ_01136	2176656	2176829	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01135	2176830	2177378	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01134	2177359	2177781	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01133	2177756	2178178	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01132	2178262	2178411	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01131	2178462	2178767	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01130	2178814	2179641	-1	hypothetical protein
2175080	2297014	121934		sspC	OGCDPPDJ_01129	2179761	2179979	1	Small, acid-soluble spore protein C
2175080	2297014	121934			OGCDPPDJ_01128	2180070	2180351	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01127	2180508	2181158	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01126	2181171	2181524	-1	hypothetical protein
2175080	2297014	121934		dfrA	OGCDPPDJ_01125	2181649	2182170	-1	Dihydrofolate reductase
2175080	2297014	121934		thyA1_1	OGCDPPDJ_01124	2182170	2183009	-1	Thymidylate synthase 1
2175080	2297014	121934			OGCDPPDJ_01123	2183066	2183482	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01122	2183525	2183740	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01121	2183724	2184029	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01120	2184342	2184959	1	hypothetical protein
2175080	2297014	121934		yncF_1	OGCDPPDJ_01119	2185053	2185481	-1	putative deoxyuridine 5'-triphosphate nucleotidohydrolase YncF
2175080	2297014	121934		trxA_2	OGCDPPDJ_01118	2185527	2185769	-1	Thioredoxin
2175080	2297014	121934		nrdF_1	OGCDPPDJ_01117	2185766	2186755	-1	Ribonucleoside-diphosphate reductase subunit beta
2175080	2297014	121934			OGCDPPDJ_01116	2186871	2187053	1	hypothetical protein
2175080	2297014	121934		nrdE_3	OGCDPPDJ_01115	2187023	2187955	-1	Ribonucleoside-diphosphate reductase subunit alpha
2175080	2297014	121934		nrdE_2	OGCDPPDJ_01114	2188257	2188694	-1	Ribonucleoside-diphosphate reductase subunit alpha
2175080	2297014	121934		nrdE_1	OGCDPPDJ_01113	2188932	2189654	-1	Ribonucleoside-diphosphate reductase subunit alpha
2175080	2297014	121934		nrdI_1	OGCDPPDJ_01112	2189617	2190012	-1	Protein NrdI
2175080	2297014	121934			OGCDPPDJ_01111	2190012	2190365	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01110	2190417	2190536	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01109	2190618	2190830	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01108	2191092	2191319	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01107	2191359	2191724	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01106	2191728	2191946	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01105	2191966	2192523	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01104	2192682	2194157	-1	hypothetical protein
2175080	2297014	121934		prs_1	OGCDPPDJ_01103	2194157	2194975	-1	Ribose-phosphate pyrophosphokinase

Island start	Island end	Length	Gene name	Gene ID	Locus	Gene start	Gene end	Strand	Product
2175080	2297014	121934			OGCDPPDJ_01102	2195120	2195467	1	hypothetical protein
2175080	2297014	121934		yorR	OGCDPPDJ_01101	2195507	2196004	-1	SPbeta prophage-derived uncharacterized protein YorR
2175080	2297014	121934			OGCDPPDJ_01100	2196004	2196159	-1	hypothetical protein
2175080	2297014	121934		yorP	OGCDPPDJ_01099	2196152	2196367	-1	SPbeta prophage-derived uncharacterized protein YorP
2175080	2297014	121934			OGCDPPDJ_01098	2196400	2196597	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01097	2196634	2196792	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01096	2196901	2197563	-1	hypothetical protein
2175080	2297014	121934		dnaE2	OGCDPPDJ_01095	2197648	2201565	-1	Error-prone DNA polymerase
2175080	2297014	121934		yorK	OGCDPPDJ_01094	2201578	2203308	-1	Putative SPbeta prophage-derived single-strand DNA-specific exonuclease YorK
2175080	2297014	121934			OGCDPPDJ_01093	2203308	2204444	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01092	2204460	2205974	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01091	2205989	2206459	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01090	2206502	2207473	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01089	2207556	2208470	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01088	2208492	2208863	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01087	2209177	2209557	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01086	2209622	2209915	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01085	2210000	2211760	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01084	2211757	2212581	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01083	2212678	2213043	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01082	2213078	2213299	-1	hypothetical protein
2175080	2297014	121934		yedK_1	OGCDPPDJ_01081	2213370	2214044	1	SOS response-associated protein YedK
2175080	2297014	121934		lig_1	OGCDPPDJ_01080	2214114	2214926	1	DNA ligase
2175080	2297014	121934			OGCDPPDJ_01079	2214992	2215405	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01078	2215458	2215571	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01077	2215568	2216374	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01076	2216375	2216875	-1	Protein SprT-like protein
2175080	2297014	121934			OGCDPPDJ_01075	2216848	2216973	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01074	2217353	2217754	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01073	2217879	2218139	1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01072	2218175	2218390	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01071	2219493	2220170	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01070	2220207	2220443	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01069	2220468	2221106	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01068	2221159	2221362	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01067	2221371	2221535	-1	hypothetical protein

Island start	Island end	Length	Gene name	Gene ID	Locus	Gene start	Gene end	Strand	Product
2175080	2297014	121934		apaH	OGCDPPDJ_01066	2221581	2223743	-1	Bis(5'-nucleosyl)-tetraphosphatase, symmetrical
2175080	2297014	121934			OGCDPPDJ_01065	2223846	2224031	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01064	2224048	2224326	-1	hypothetical protein
2175080	2297014	121934		yopX	OGCDPPDJ_01063	2224323	2224718	-1	SPbeta prophage-derived uncharacterized protein YopX
2175080	2297014	121934			OGCDPPDJ_01062	2224718	2225368	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01061	2225736	2225930	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01060	2226042	2226239	-1	hypothetical protein
2175080	2297014	121934		yopT	OGCDPPDJ_01059	2226309	2226527	-1	SPbeta prophage-derived uncharacterized protein YopT
2175080	2297014	121934			OGCDPPDJ_01058	2226710	2226934	1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01057	2227123	2228100	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01056	2228124	2229506	-1	hypothetical protein
2175080	2297014	121934		xerC_2	OGCDPPDJ_01055	2229613	2230689	-1	Tyrosine recombinase XerC
2175080	2297014	121934			OGCDPPDJ_01054	2230679	2230891	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01053	2230940	2231257	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01052	2231260	2231460	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01051	2231789	2231914	-1	hypothetical protein
2175080	2297014	121934		yopK	OGCDPPDJ_01050	2231928	2233088	-1	SPbeta prophage-derived uncharacterized protein YopK
2175080	2297014	121934			OGCDPPDJ_01049	2233259	2233828	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01048	2233850	2234206	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01047	2234237	2234710	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01046	2234784	2235272	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01045	2235361	2235795	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01044	2235831	2236046	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01043	2236103	2236300	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01042	2236622	2236846	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01041	2236978	2237574	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01040	2237610	2237873	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01039	2237896	2238561	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01038	2238644	2238877	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01037	2238950	2239171	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01036	2239262	2239699	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01035	2239790	2240236	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01034	2240271	2241215	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01033	2241270	2241494	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01032	2241572	2241736	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01031	2241733	2241975	-1	hypothetical protein

Island start	Island end	Length	Gene name	Gene ID	Locus	Gene start	Gene end	Strand	Product
2175080	2297014	121934			OGCDPPDJ_01030	2242000	2242224	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01029	2242217	2242348	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01028	2242368	2242703	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01027	2243046	2244263	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01026	2244282	2245127	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01025	2245273	2245434	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01024	2246266	2246448	1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01023	2246960	2247154	1	hypothetical protein
2175080	2297014	121934		yonO	OGCDPPDJ_01022	2247194	2249701	1	DNA-directed RNA polymerase YonO
2175080	2297014	121934		hup	OGCDPPDJ_01021	2249964	2250242	1	DNA-binding protein HU
2175080	2297014	121934			OGCDPPDJ_01020	2251938	2252123	1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01019	2252135	2253334	1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01018	2253348	2253803	1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01017	2253929	2254924	1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01016	2254924	2256672	1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01015	2256696	2258234	1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01014	2258253	2259401	1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01013	2259438	2259902	1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01012	2259931	2261034	1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01011	2261085	2261558	1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01010	2261570	2261956	1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01009	2261966	2262628	1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01008	2262625	2263131	1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01007	2263128	2263838	1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01006	2263881	2264678	1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01005	2264798	2265307	1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01004	2265307	2265534	1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01003	2265598	2265954	1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01002	2265954	2266982	1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01001	2266996	2267349	1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_01000	2267342	2267539	1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_00999	2267599	2268099	1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_00998	2268074	2268502	1	hypothetical protein
2175080	2297014	121934		xerC_1	OGCDPPDJ_00997	2268516	2269523	1	Tyrosine recombinase XerC
2175080	2297014	121934			OGCDPPDJ_00996	2269665	2270132	1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_00995	2270283	2270477	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_00994	2270500	2271333	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_00993	2271693	2272376	1	hypothetical protein

Island start	Island end	Length	Gene name	Gene ID	Locus	Gene start	Gene end	Strand	Product
2175080	2297014	121934			OGCDPPDJ_00992	2272430	2279320	1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_00991	2279361	2280122	1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_00990	2280135	2282777	1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_00989	2282783	2283610	1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_00988	2283623	2285497	1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_00987	2285651	2286718	1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_00986	2286834	2287226	1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_00985	2287247	2287498	1	hypothetical protein
2175080	2297014	121934		rapA_3	OGCDPPDJ_00984	2287626	2288762	1	Response regulator aspartate phosphatase A
2175080	2297014	121934			OGCDPPDJ_00983	2288752	2288928	1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_00982	2288970	2289188	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_00981	2289265	2289585	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_00980	2289753	2290733	1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_00979	2291280	2291861	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_00978	2291917	2292375	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_00977	2292388	2294274	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_00976	2294570	2294944	-1	hypothetical protein
2175080	2297014	121934			OGCDPPDJ_00975	2295032	2295283	-1	hypothetical protein
2175080	2297014	121934		yokF	OGCDPPDJ_00974	2295466	2296356	1	SPbeta prophage-derived endonuclease YokF
2175080	2297014	121934			OGCDPPDJ_00973	2296481	2297014	-1	hypothetical protein
2207556	2213299	5743			OGCDPPDJ_01089	2207556	2208470	-1	hypothetical protein
2207556	2213299	5743			OGCDPPDJ_01088	2208492	2208863	-1	hypothetical protein
2207556	2213299	5743			OGCDPPDJ_01087	2209177	2209557	-1	hypothetical protein
2207556	2213299	5743			OGCDPPDJ_01086	2209622	2209915	-1	hypothetical protein
2207556	2213299	5743			OGCDPPDJ_01085	2210000	2211760	-1	hypothetical protein
2207556	2213299	5743			OGCDPPDJ_01084	2211757	2212581	-1	hypothetical protein
2207556	2213299	5743			OGCDPPDJ_01083	2212678	2213043	-1	hypothetical protein
2207556	2213299	5743			OGCDPPDJ_01082	2213078	2213299	-1	hypothetical protein
2226309	2231460	5151		yopT	OGCDPPDJ_01059	2226309	2226527	-1	SPbeta prophage-derived uncharacterized protein YopT
2226309	2231460	5151			OGCDPPDJ_01058	2226710	2226934	1	hypothetical protein
2226309	2231460	5151			OGCDPPDJ_01057	2227123	2228100	-1	hypothetical protein
2226309	2231460	5151			OGCDPPDJ_01056	2228124	2229506	-1	hypothetical protein
2226309	2231460	5151		xerC_2	OGCDPPDJ_01055	2229613	2230689	-1	Tyrosine recombinase XerC
2226309	2231460	5151			OGCDPPDJ_01054	2230679	2230891	-1	hypothetical protein
2226309	2231460	5151			OGCDPPDJ_01053	2230940	2231257	-1	hypothetical protein
2226309	2231460	5151			OGCDPPDJ_01052	2231260	2231460	-1	hypothetical protein

Island start	Island end	Length	Gene name	Gene ID	Locus	Gene start	Gene end	Strand	Product
2231928	2239171	7243		yopK	OGCDPPDJ_01050	2231928	2233088	-1	SPbeta prophage-derived uncharacterized protein YopK
2231928	2239171	7243			OGCDPPDJ_01049	2233259	2233828	-1	hypothetical protein
2231928	2239171	7243			OGCDPPDJ_01048	2233850	2234206	-1	hypothetical protein
2231928	2239171	7243			OGCDPPDJ_01047	2234237	2234710	-1	hypothetical protein
2231928	2239171	7243			OGCDPPDJ_01046	2234784	2235272	-1	hypothetical protein
2231928	2239171	7243			OGCDPPDJ_01045	2235361	2235795	-1	hypothetical protein
2231928	2239171	7243			OGCDPPDJ_01044	2235831	2236046	-1	hypothetical protein
2231928	2239171	7243			OGCDPPDJ_01043	2236103	2236300	-1	hypothetical protein
2231928	2239171	7243			OGCDPPDJ_01042	2236622	2236846	-1	hypothetical protein
2231928	2239171	7243			OGCDPPDJ_01041	2236978	2237574	-1	hypothetical protein
2231928	2239171	7243			OGCDPPDJ_01040	2237610	2237873	-1	hypothetical protein
2231928	2239171	7243			OGCDPPDJ_01039	2237896	2238561	-1	hypothetical protein
2231928	2239171	7243			OGCDPPDJ_01038	2238644	2238877	-1	hypothetical protein
2231928	2239171	7243			OGCDPPDJ_01037	2238950	2239171	-1	hypothetical protein
2239790	2245127	5337			OGCDPPDJ_01035	2239790	2240236	-1	hypothetical protein
2239790	2245127	5337			OGCDPPDJ_01034	2240271	2241215	-1	hypothetical protein
2239790	2245127	5337			OGCDPPDJ_01033	2241270	2241494	-1	hypothetical protein
2239790	2245127	5337			OGCDPPDJ_01032	2241572	2241736	-1	hypothetical protein
2239790	2245127	5337			OGCDPPDJ_01031	2241733	2241975	-1	hypothetical protein
2239790	2245127	5337			OGCDPPDJ_01030	2242000	2242224	-1	hypothetical protein
2239790	2245127	5337			OGCDPPDJ_01029	2242217	2242348	-1	hypothetical protein
2239790	2245127	5337			OGCDPPDJ_01028	2242368	2242703	-1	hypothetical protein
2239790	2245127	5337			OGCDPPDJ_01027	2243046	2244263	-1	hypothetical protein
2239790	2245127	5337			OGCDPPDJ_01026	2244282	2245127	-1	hypothetical protein
2279361	2285497	6136			OGCDPPDJ_00991	2279361	2280122	1	hypothetical protein
2279361	2285497	6136			OGCDPPDJ_00990	2280135	2282777	1	hypothetical protein
2279361	2285497	6136			OGCDPPDJ_00989	2282783	2283610	1	hypothetical protein
2279361	2285497	6136			OGCDPPDJ_00988	2283623	2285497	1	hypothetical protein
2288752	2295283	6531		rapA_3	OGCDPPDJ_00984	2287626	2288762	1	Response regulator aspartate phosphatase A
2288752	2295283	6531			OGCDPPDJ_00983	2288752	2288928	1	hypothetical protein
2288752	2295283	6531			OGCDPPDJ_00982	2288970	2289188	-1	hypothetical protein
2288752	2295283	6531			OGCDPPDJ_00981	2289265	2289585	-1	hypothetical protein
2288752	2295283	6531			OGCDPPDJ_00980	2289753	2290733	1	hypothetical protein
2288752	2295283	6531			OGCDPPDJ_00979	2291280	2291861	-1	hypothetical protein
2288752	2295283	6531			OGCDPPDJ_00978	2291917	2292375	-1	hypothetical protein
2288752	2295283	6531			OGCDPPDJ_00977	2292388	2294274	-1	hypothetical protein
2288752	2295283	6531			OGCDPPDJ_00976	2294570	2294944	-1	hypothetical protein
2288752	2295283	6531			OGCDPPDJ_00975	2295032	2295283	-1	hypothetical protein

Island start	Island end	Length	Gene name	Gene ID	Locus	Gene start	Gene end	Strand	Product
2672061	2722771	50710		acr3	OGCDPPDJ_00542	2672061	2673101	-1	Arsenical-resistance protein Acr3
2672061	2722771	50710		cadI	OGCDPPDJ_00541	2673124	2673564	-1	Cadmium-induced protein CadI
2672061	2722771	50710		arsR	OGCDPPDJ_00540	2673625	2673942	-1	Arsenical resistance operon repressor
2672061	2722771	50710			OGCDPPDJ_00539	2674313	2675077	-1	hypothetical protein
2672061	2722771	50710			OGCDPPDJ_00538	2675391	2675861	-1	hypothetical protein
2672061	2722771	50710			OGCDPPDJ_00537	2675890	2676312	-1	hypothetical protein
2672061	2722771	50710			OGCDPPDJ_00536	2676387	2677253	-1	hypothetical protein
2672061	2722771	50710		ttgR	OGCDPPDJ_00535	2677795	2678388	-1	HTH-type transcriptional regulator TtgR
2672061	2722771	50710		yhfP_1	OGCDPPDJ_00534	2678546	2679541	1	Putative quinone oxidoreductase YhfP
2672061	2722771	50710			OGCDPPDJ_00533	2680072	2680947	-1	hypothetical protein
2672061	2722771	50710			OGCDPPDJ_00532	2680998	2681345	-1	hypothetical protein
2672061	2722771	50710			OGCDPPDJ_00531	2681422	2682087	-1	hypothetical protein
2672061	2722771	50710		rapA_1	OGCDPPDJ_00530	2682386	2683522	1	Response regulator aspartate phosphatase A
2672061	2722771	50710			OGCDPPDJ_00529	2683512	2683646	1	hypothetical protein
2672061	2722771	50710			OGCDPPDJ_00528	2683756	2684169	1	hypothetical protein
2672061	2722771	50710		yqcG_1	OGCDPPDJ_00527	2684285	2685880	1	Ribonuclease YqcG
2672061	2722771	50710		yqcF	OGCDPPDJ_00526	2685895	2686473	1	Antitoxin YqcF
2672061	2722771	50710			OGCDPPDJ_00525	2686591	2686737	1	hypothetical protein
2672061	2722771	50710			OGCDPPDJ_00524	2686734	2687096	-1	hypothetical protein
2672061	2722771	50710			OGCDPPDJ_00523	2687112	2687591	-1	hypothetical protein
2672061	2722771	50710		cwlA_1	OGCDPPDJ_00522	2687756	2688574	-1	N-acetylmuramoyl-L-alanine amidase CwlA
2672061	2722771	50710			OGCDPPDJ_00521	2688619	2689041	-1	hypothetical protein
2672061	2722771	50710			OGCDPPDJ_00520	2689086	2689979	-1	hypothetical protein
2672061	2722771	50710			OGCDPPDJ_00519	2690066	2690230	-1	hypothetical protein
2672061	2722771	50710		xkdW_1	OGCDPPDJ_00518	2690227	2690568	-1	Phage-like element PBSX protein XkdW
2672061	2722771	50710			OGCDPPDJ_00517	2690578	2691672	-1	hypothetical protein
2672061	2722771	50710			OGCDPPDJ_00516	2691676	2691948	-1	hypothetical protein
2672061	2722771	50710			OGCDPPDJ_00515	2691945	2692523	-1	hypothetical protein
2672061	2722771	50710			OGCDPPDJ_00514	2692507	2693553	-1	hypothetical protein
2672061	2722771	50710			OGCDPPDJ_00513	2693546	2693971	-1	hypothetical protein
2672061	2722771	50710		ygeA	OGCDPPDJ_00512	2694073	2694771	-1	L-aspartate/glutamate-specific racemase
2672061	2722771	50710			OGCDPPDJ_00511	2694861	2695127	-1	hypothetical protein
2672061	2722771	50710			OGCDPPDJ_00510	2695124	2696104	-1	hypothetical protein
2672061	2722771	50710			OGCDPPDJ_00509	2696117	2696824	-1	hypothetical protein

Island start	Island end	Length	Gene name	Gene ID	Locus	Gene start	Gene end	Strand	Product
2672061	2722771	50710		yqbO_1	OGCDPPDJ_00508	2696769	2701526	-1	putative protein YqbO
2672061	2722771	50710			OGCDPPDJ_00507	2701529	2701708	-1	hypothetical protein
2672061	2722771	50710		yqbN_1	OGCDPPDJ_00506	2701708	2702157	-1	putative protein YqbN
2672061	2722771	50710		xkdM_1	OGCDPPDJ_00505	2702691	2703134	-1	Phage-like element PBSX protein XkdM
2672061	2722771	50710			OGCDPPDJ_00504	2703137	2704537	-1	hypothetical protein
2672061	2722771	50710			OGCDPPDJ_00503	2704538	2704726	-1	hypothetical protein
2672061	2722771	50710			OGCDPPDJ_00502	2704726	2705163	-1	hypothetical protein
2672061	2722771	50710			OGCDPPDJ_00501	2705176	2705679	-1	hypothetical protein
2672061	2722771	50710		xkdH_1	OGCDPPDJ_00500	2705676	2706038	-1	Phage-like element PBSX protein XkdH
2672061	2722771	50710		yqbG_1	OGCDPPDJ_00499	2706035	2706430	-1	putative protein YqbG
2672061	2722771	50710		yqbF	OGCDPPDJ_00498	2706434	2706745	-1	putative protein YqbF
2672061	2722771	50710		xkdG_1	OGCDPPDJ_00497	2706756	2707691	-1	Phage-like element PBSX protein XkdG
2672061	2722771	50710		yqbD_1	OGCDPPDJ_00496	2707710	2708684	-1	putative protein YqbD
2672061	2722771	50710			OGCDPPDJ_00495	2708771	2709547	-1	hypothetical protein
2672061	2722771	50710			OGCDPPDJ_00494	2709603	2710514	-1	hypothetical protein
2672061	2722771	50710			OGCDPPDJ_00493	2710511	2712043	-1	hypothetical protein
2672061	2722771	50710			OGCDPPDJ_00492	2712047	2713342	-1	hypothetical protein
2672061	2722771	50710			OGCDPPDJ_00491	2713335	2714054	-1	hypothetical protein
2672061	2722771	50710			OGCDPPDJ_00490	2714578	2714766	-1	hypothetical protein
2672061	2722771	50710			OGCDPPDJ_00489	2714911	2715366	-1	hypothetical protein
2672061	2722771	50710			OGCDPPDJ_00488	2715457	2716110	-1	hypothetical protein
2672061	2722771	50710			OGCDPPDJ_00487	2716177	2716383	-1	hypothetical protein
2672061	2722771	50710			OGCDPPDJ_00486	2716465	2716893	-1	hypothetical protein
2672061	2722771	50710			OGCDPPDJ_00485	2716989	2717132	-1	hypothetical protein
2672061	2722771	50710			OGCDPPDJ_00484	2717129	2717989	-1	hypothetical protein
2672061	2722771	50710			OGCDPPDJ_00483	2717952	2718653	-1	hypothetical protein
2672061	2722771	50710		recT	OGCDPPDJ_00482	2718706	2719560	-1	Protein RecT
2672061	2722771	50710			OGCDPPDJ_00481	2719563	2720519	-1	hypothetical protein
2672061	2722771	50710		yqaI	OGCDPPDJ_00480	2720623	2720817	-1	putative protein YqaI
2672061	2722771	50710			OGCDPPDJ_00479	2720777	2720950	-1	hypothetical protein
2672061	2722771	50710			OGCDPPDJ_00478	2720947	2721204	-1	hypothetical protein
2672061	2722771	50710			OGCDPPDJ_00477	2721201	2721770	-1	hypothetical protein
2672061	2722771	50710			OGCDPPDJ_00476	2721844	2722017	-1	hypothetical protein
2672061	2722771	50710			OGCDPPDJ_00475	2722014	2722244	-1	hypothetical protein
2672061	2722771	50710		xre_1	OGCDPPDJ_00474	2722421	2722771	1	HTH-type transcriptional regulator Xre
2673124	2678388	5264		cadI	OGCDPPDJ_00541	2673124	2673564	-1	Cadmium-induced protein CadI

Island start	Island end	Length	Gene name	Gene ID	Locus	Gene start	Gene end	Strand	Product
2673124	2678388	5264		arsR	OGCDPPDJ_00540	2673625	2673942	-1	Arsenical resistance operon repressor
2673124	2678388	5264			OGCDPPDJ_00539	2674313	2675077	-1	hypothetical protein
2673124	2678388	5264			OGCDPPDJ_00538	2675391	2675861	-1	hypothetical protein
2673124	2678388	5264			OGCDPPDJ_00537	2675890	2676312	-1	hypothetical protein
2673124	2678388	5264			OGCDPPDJ_00536	2676387	2677253	-1	hypothetical protein
2673124	2678388	5264		ttgR	OGCDPPDJ_00535	2677795	2678388	-1	HTH-type transcriptional regulator TtgR
2722014	2726532	4518			OGCDPPDJ_00476	2721844	2722017	-1	hypothetical protein
2722014	2726532	4518			OGCDPPDJ_00475	2722014	2722244	-1	hypothetical protein
2722014	2726532	4518		xre_1	OGCDPPDJ_00474	2722421	2722771	1	HTH-type transcriptional regulator Xre
2722014	2726532	4518			OGCDPPDJ_00473	2723038	2723205	-1	hypothetical protein
2722014	2726532	4518			OGCDPPDJ_00472	2723561	2724097	-1	hypothetical protein
2722014	2726532	4518			OGCDPPDJ_00471	2724366	2724884	1	hypothetical protein
2722014	2726532	4518		sigK_1	OGCDPPDJ_00470	2724884	2725282	1	RNA polymerase sigma-K factor
2722014	2726532	4518		tetR_1	OGCDPPDJ_00469	2725852	2726532	-1	Tetracycline repressor protein class B from transposon Tn10
2740931	2745918	4987			OGCDPPDJ_00454	2740931	2741074	-1	hypothetical protein
2740931	2745918	4987			OGCDPPDJ_00453	2741091	2742056	-1	hypothetical protein
2740931	2745918	4987		gltR_2	OGCDPPDJ_00452	2742182	2743048	1	HTH-type transcriptional regulator GltR
2740931	2745918	4987		czcO	OGCDPPDJ_00451	2743169	2744206	-1	putative oxidoreductase CzcO
2740931	2745918	4987		czcD	OGCDPPDJ_00450	2744294	2745241	-1	Cadmium, cobalt and zinc/H(+)-K(+) antiporter
2740931	2745918	4987			OGCDPPDJ_00449	2745529	2745918	-1	hypothetical protein
3502140	3506605	4465			OGCDPPDJ_01744	3502140	3502970	-1	hypothetical protein
3502140	3506605	4465			OGCDPPDJ_01745	3502954	3503829	-1	hypothetical protein
3502140	3506605	4465			OGCDPPDJ_01746	3503822	3504040	-1	hypothetical protein
3502140	3506605	4465			OGCDPPDJ_01747	3504063	3504254	-1	hypothetical protein
3502140	3506605	4465			OGCDPPDJ_01748	3504296	3504502	-1	hypothetical protein
3502140	3506605	4465			OGCDPPDJ_01749	3504667	3505065	1	hypothetical protein
3502140	3506605	4465			OGCDPPDJ_01750	3505062	3505154	-1	hypothetical protein
3502140	3506605	4465			OGCDPPDJ_01751	3505345	3505476	-1	hypothetical protein
3502140	3506605	4465			OGCDPPDJ_01752	3505496	3506605	-1	hypothetical protein
3729898	3734132	4234			OGCDPPDJ_01971	3729898	3731211	1	hypothetical protein
3729898	3734132	4234			OGCDPPDJ_01972	3731162	3731350	1	hypothetical protein
3729898	3734132	4234			OGCDPPDJ_01973	3731556	3734132	-1	hypothetical protein
4105313	4109740	4427		uvrB_2	OGCDPPDJ_02350	4105313	4108447	1	UvrABC system protein B
4105313	4109740	4427			OGCDPPDJ_02351	4108475	4109332	-1	hypothetical protein
4105313	4109740	4427		nudG	OGCDPPDJ_02352	4109333	4109740	-1	CTP pyrophosphohydrolase

Island start	Island end	Length	Gene name	Gene ID	Locus	Gene start	Gene end	Strand	Product
4199284	4206479	7195			OGCDPPDJ_02438	4199284	4199442	1	hypothetical protein
4199284	4206479	7195		yydJ	OGCDPPDJ_02439	4199576	4200298	-1	putative peptide export permease protein YydJ
4199284	4206479	7195		yydI	OGCDPPDJ_02440	4200319	4200948	-1	putative peptide export ATP-binding protein YydI
4199284	4206479	7195		fbp	OGCDPPDJ_02441	4201430	4203355	1	Fructose-1,6-bisphosphatase class 3
4199284	4206479	7195			OGCDPPDJ_02442	4203456	4206479	-1	hypothetical protein
4209002	4217871	8869			OGCDPPDJ_02445	4209002	4210534	-1	hypothetical protein
4209002	4217871	8869			OGCDPPDJ_02446	4210635	4211927	-1	hypothetical protein
4209002	4217871	8869			OGCDPPDJ_02447	4212652	4213338	1	hypothetical protein
4209002	4217871	8869		rlmH	OGCDPPDJ_02448	4213518	4213997	-1	Ribosomal RNA large subunit methyltransferase H
4209002	4217871	8869			OGCDPPDJ_02449	4214078	4214248	-1	hypothetical protein
4209002	4217871	8869			OGCDPPDJ_02450	4214433	4214846	1	hypothetical protein
4209002	4217871	8869		fdhA	OGCDPPDJ_02451	4214880	4216091	-1	Glutathione-independent formaldehyde dehydrogenase
4209002	4217871	8869			OGCDPPDJ_02452	4216169	4216339	-1	hypothetical protein
4209002	4217871	8869			OGCDPPDJ_02453	4216444	4216692	-1	hypothetical protein
4209002	4217871	8869			OGCDPPDJ_02454	4216708	4217871	-1	hypothetical protein
4260056	4269218	9162			OGCDPPDJ_02502	4260056	4260535	1	hypothetical protein
4260056	4269218	9162		gloB_4	OGCDPPDJ_02503	4260710	4261372	-1	Hydroxyacylglutathione hydrolase
4260056	4269218	9162			OGCDPPDJ_02504	4261518	4261970	-1	hypothetical protein
4260056	4269218	9162		yjcF_2	OGCDPPDJ_02505	4262090	4262530	1	putative N-acetyltransferase YjcF
4260056	4269218	9162			OGCDPPDJ_02506	4262532	4263134	1	hypothetical protein
4260056	4269218	9162		satA	OGCDPPDJ_02507	4263202	4263723	-1	Streptothricin acetyltransferase A
4260056	4269218	9162			OGCDPPDJ_02508	4264134	4264490	1	hypothetical protein
4260056	4269218	9162		yyaP	OGCDPPDJ_02509	4264650	4265216	1	putative protein YyaP
4260056	4269218	9162		tet(L)	OGCDPPDJ_02510	4265733	4267109	-1	tetracycline efflux MFS transporter Tet(L)
4260056	4269218	9162			OGCDPPDJ_02511	4267457	4267627	1	hypothetical protein
4260056	4269218	9162			OGCDPPDJ_02512	4267710	4269218	1	hypothetical protein
4267710	4271748	4038			OGCDPPDJ_02512	4267710	4269218	1	hypothetical protein
4267710	4271748	4038			OGCDPPDJ_02513	4269723	4270511	-1	hypothetical protein
4267710	4271748	4038		yidD	OGCDPPDJ_02514	4270508	4270753	-1	Putative membrane protein insertion efficiency factor
4267710	4271748	4038		hemB_2	OGCDPPDJ_02515	4270750	4271748	-1	Delta-aminolevulinic acid dehydratase
4342769	4411758	68989		Hgd	OGCDPPDJ_04480	4342769	4343053	1	2-(hydroxymethyl)glutarate dehydrogenase
4342769	4411758	68989			OGCDPPDJ_04479	4376199	4376540	-1	hypothetical protein

Island start	Island end	Length	Gene name	Gene ID	Locus	Gene start	Gene end	Strand	Product
4342769	4411758	68989			OGCDPPDJ_04487	4411480	4411758	-1	hypothetical protein

Đánh giá

- Dự đoán được 30 Genome Island
- Đa số có kích thước nhỏ (4-9 kb), do SIGI-HMM dự đoán
- Có 4 vùng lớn (≥15 kb), do IslandPath-DIMOB dự đoán
- Cần ưu tiên vào 4 vùng có kích thước lớn vì độ tin cậy cao

Start	End	Length	Số lượng genes	ID	%GC	Gene mobility + HGT markers	Contigs
1191680	1270652	78972	116	GI_1	46,09%	Có gene Tyrosine recombinase XerC	3
2148724	2163692	14968	22	GI_2	35,52%	Có gene Tyrosine recombinase XerD	1
2175080	2297014	121934	168	GI_3	34,83%	Có gene SPbeta prophage-derived uncharacterized protein YopX, SPbeta prophage-derived uncharacterized protein YopT, Tyrosine recombinase XerC, Tyrosine recombinase XerC	1
2672061	2722771	50710	69	GI_4	38,56%	Có gene Phage-like element PBSX protein XkdG, Phage-like element PBSX protein XkdH, Phage-like element PBSX protein XkdM	1

a/ Kiểm tra GC content và GC skew

- So sánh: GC% của GI với GC% toàn hệ gene. Nếu lệch ≥2–5% là tín hiệu HGT tốt
- GI có điểm gãy GC skew
- Kết quả: GI_1 (46,09%), GI_2 (35,52%), GI_3 (34,83%), GI_4 (38,56%) và GC% toàn hệ gene là 43,52%. 4 vùng đều đủ tiêu chuẩn
- 4 vùng đều có %GC lệch 3-5% so với toàn genome (43,52%)

b/ Kiểm tra sự hiện diện của các gene mobility (chạy bằng mobileOG-db, pident >= 90%), bao gồm:

- Integrase / Recombinase
- Transposase / IS element
- Phage-related proteins
- Tra / T4SS / conjugation proteins
- Plasmid maintenance proteins
- Hypothetical proteins (tỷ lệ cao bất thường)
- Kết quả: 4 vùng đều có các gene liên quan recombinase, prophage-derived protein, phage-like protein, ...
- Vùng GI_1 có 4 trình tự (3 phage,1 transfer)
- Không có gene nào ở GI_2
- Vùng GI_3 có khoảng 13 trình tự được xác định
- Vùng GI_4 có khoảng 15 trình tự được xác định (12 trình tự thuộc phage, 2 trình tự thuộc replication/recombination/repair và 1 trình tự thuộc integration/excision)

c/ Kiểm tra vị trí gắn với tRNA

- Rất nhiều GI (đặc biệt PAI) chèn vào: tRNA genes, mRNA
- Kết quả: Chỉ có vùng GI_1 nằm ở contigs 3, có vị trí 695003-695077 (03275)

d/ Kiểm tra bằng các công cụ độc lập

- **AlienHunter**: Dự đoán các HCT events (Kiểm tra contig 1 và 3). Đối với contig 1; có 2 vùng với vị trí 1023670-1041383 (trùng với vị trí GI_2, score=0.691), 892936-1018093 (trùng với vị trí GI_3, score=0.6). Các vị trí còn lại không phù hợp, có score thấp. Đối với contig 3; có 3 vùng với vị trí 1-10000 (score=0.764), 615437-695036 (trùng với vị trí GI_1, score=0.733), 826928-832780 (score=0.992)

- **PHASTEST**: Phát hiện và chú giải các vùng prophage (Kiểm tra contig 1 và 3). Đối với contig 1; có 2 vùng được xác định ở vị trí 479706-508763 (trùng với vị trí GI_4, completeness=intact) và vị trí 892932-1018193 (trùng với vị trí GI_3, completeness=intact). Đối với contig 3; có 2 vùng được xác định ở vị trí 537998-571613 (completeness=intact) và 622175-694845 (trùng với GI_1, completeness=intact)

e/ Tổng kết các GI tiềm năng

ID	Contig	Bắt đầu	Kết thúc	Độ dài	%GC	GC skew	mobileOG-db (pident >= 90%)	Gắn với tRNA	AlienHunter
GI_1	3	617456	696437	78971	46,09%	Có đổi dấu	Có 4 gene	Không	Có, score=0,733
GI_2	1	1029356	1044324	14968	35,52%	Không đổi dấu	Không có gene	Không	Có, score=0,691
GI_3	1	896034	1017968	121934	34,83%	Có đổi dấu	Có 13 gene	Có	Có, score=0,6
GI_4	1	470277	520987	50710	38,56%	Không đổi dấu	Có 15 gene	Không	Không có

- Kết quả: Xác nhận 3 vùng GI_1, GI_3 và GI_4 là các vùng Genome Island tiềm năng

Pathogenicity Island

Giới thiệu về PAI

- Là 1 phân nhóm đặc biệt của GI
- Đóng vai trò trực tiếp/gián tiếp trong khả năng gây bệnh

Đặc điểm của PAI

- Có kích thước lớn (20-200kB)
- Chứa toxin, adhension, invasion protein, secrection system
- Là một cụm gene (khác với các gene độc lực riêng lẻ) được thu nhận như một đơn vị tiến hóa

Phân tích PAI (GI + Viruslence genes)

- Xác định chức năng của các gene trong GI tiềm năng
- Vùng nào có các gene liên quan đến độc lực như: Secretion systems, Toxin/effector, Adhesion & invasion, Iron acquisition (rất phổ biến), Immune evasion
- Dùng BLAST/HMM với database chuẩn (VFDB)
- Kết quả: Sử dụng VFAnalyzer kiểm tra các gene độc lực nhưng không có xuất hiện các gene độc lực ở vùng GI tiềm năng => Không có PAI